Disjoint Set Union

September 7, 2026

A **(ID 2200110)**

B **(ID 2200110)**

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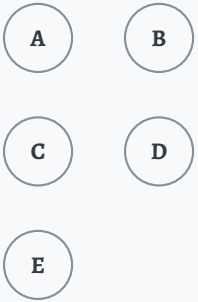
Department of Computer Science and Engineering

Outline

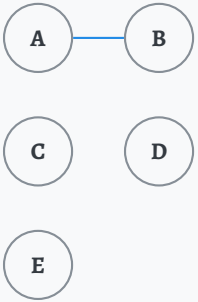
Introduction

A Familiar Problem



- Suppose there are five students in a classroom.
- At first, none of them know one another.

A Familiar Problem



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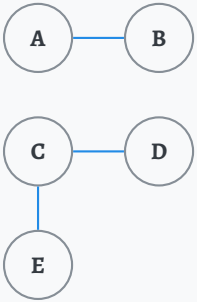

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- A and B become friends.
- Separately, C become friends with D and E.

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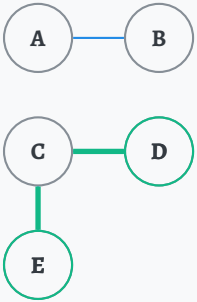

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- Are D and E friends? Yes — indirectly, through C, they belong to same friend circle.

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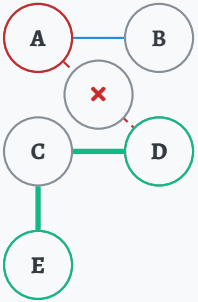

- Suppose there are five students in a classroom.
- At first, none of them know one another.
- A and B become friends.
- Separately, C become friends with D and E.
- Are D and E friends? Yes — indirectly, through C, they belong to same friend circle.
- What about A and D? No — they belong to two different circles entirely.

Larger Networks

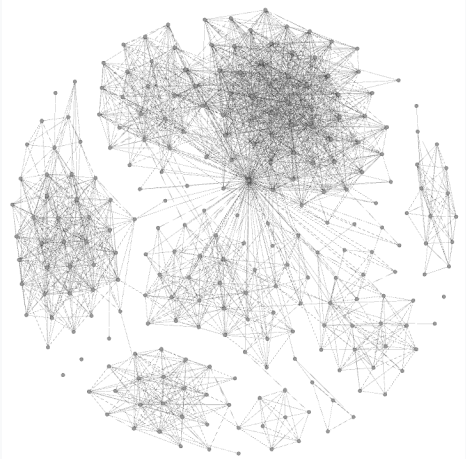

Figure: *Social Media Network Visualization*[a]

- Consider a large social network, or equally, a large computer network, with countless members.

[a] Pinterest: Visualise Your Facebook Friend Network

Larger Networks

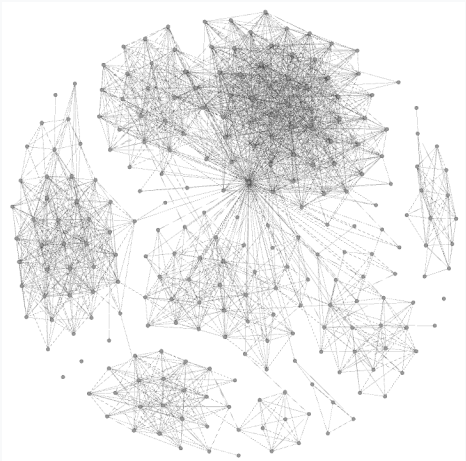

Figure: *Social Media Network Visualization*[a]

- Consider a large social network, or equally, a large computer network, with countless members.
- We need to answer **efficiently**: are any two given members connected?

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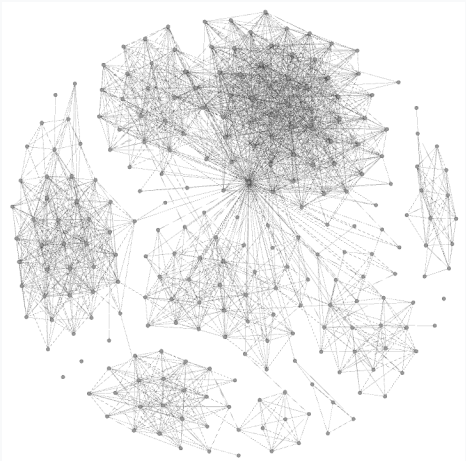

Figure: *Social Media Network Visualization*[a]

- Consider a large social network, or equally, a large computer network, with countless members.
- We need to answer **efficiently**: are any two given members connected?

This is exactly what *Disjoint Set Union* answers.

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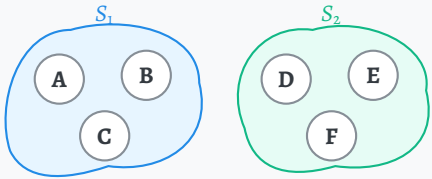
What Is a Disjoint Set Union?



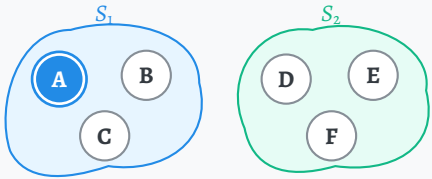
Definition

- Sets are disjoint if they share no elements:

$$S_i \cap S_j = \varnothing \quad \text{for } i \neq j.$$

- **Disjoint Set Union** or **DSU** are a collection of such disjoint groups.

What Is a Disjoint Set Union?



Here,
- A is the representative of S_1

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Representative (“leader”)
- One chosen element identifies the entire set.

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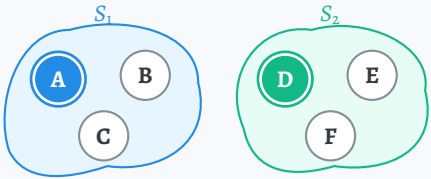

Here,
- A is the representative of S_1
- D is the representative of S_2

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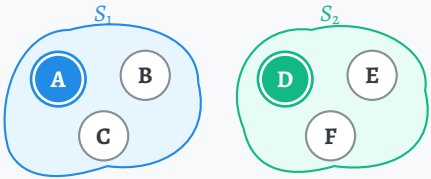

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Same representative $\iff$ same set.

The Interface

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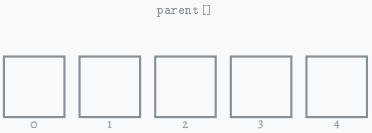
Bacause of the two main operations, `union` and `find`, this structure is also called **Union–Find**.

Naive Implementation

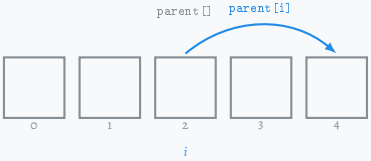
The Setup

- Maintain an array `parent[]`



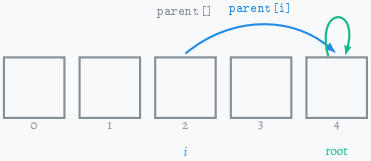
The Setup

- Maintain an array `parent[]`
- `parent[i]` stores the index of element *i*'s **immediate parent** — not necessarily its representative.



The Setup

- Maintain an array `parent[]`
- `parent[i]` stores the index of element *i*'s **immediate parent** — not necessarily its representative.
- Following parents eventually reaches the **root**, which is exactly the representative: a root points to itself, `parent[root] = root`.



make_set

Pseudocode

```
1 make_set(node):
2     parent[node] ← node
```

Call once for each new element.

make_set

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Call once for each new element.

Parent array

index	0	1	2	3	4	5	6
parent	–	–	–	–	–	–	–

– means not initialized yet.

DSU Forest

No sets created yet.

make_set

Pseudocode

```
1 make_set(node):
2     parent[node] ← node
```

Call once for each new element.

Operations

```
make_set(0):parent[0] = 0
```

Parent array

index	0	1	2	3	4	5	6
parent	0	–	–	–	–	–	–

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1 make_set(node):
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Call once for each new element.

Operations

```
make_set(0):parent[0] = 0
make_set(1):parent[1] = 1
```

Parent array

index	0	1	2	3	4	5	6
parent	0	1	–	–	–	–	–

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Operations

```
make_set(0):parent[0] = 0
make_set(1):parent[1] = 1
make_set(2):parent[2] = 2
```

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	–	–	–	–

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Operations

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make_set(0):parent[0] = 0
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make_set(3):parent[3] = 3
```

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parent	0	1	2	3	–	–	–

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make_set(3):parent[3] = 3
make_set(4):parent[4] = 4
```

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	–	–

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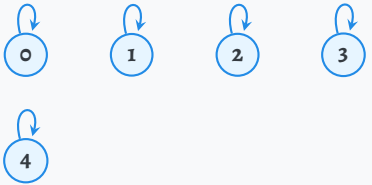

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make_set(2):parent[2] = 2
make_set(3):parent[3] = 3
make_set(4):parent[4] = 4
make_set(5):parent[5] = 5
```

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index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	–

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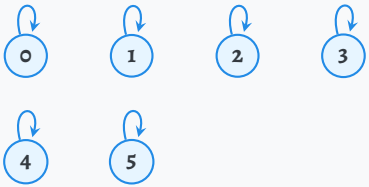

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make_set(4):parent[4] = 4
make_set(5):parent[5] = 5
make_set(6):parent[6] = 6
```

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	6

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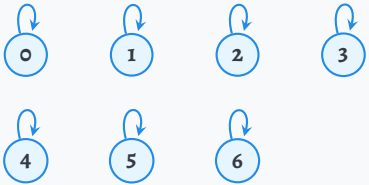
DSU Forest



`find`: Follow Parents to the Root

Pseudocode

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
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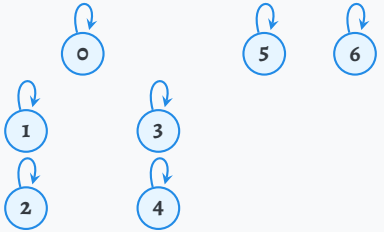

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```
1 find(node):
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4     return find(parent[node])
```

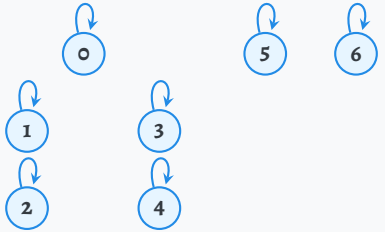
Operations

`find(4)`

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	6

DSU Forest



`find`: Follow Parents to the Root

Pseudocode

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
```

Operations

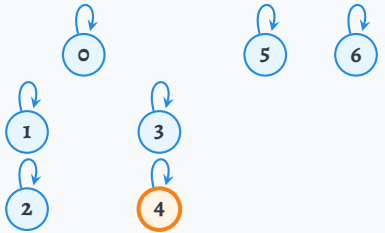
`find(4)`

- Read `parent[4] = 4`.

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	6

DSU Forest



`find`: Follow Parents to the Root

Pseudocode

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
```

Operations

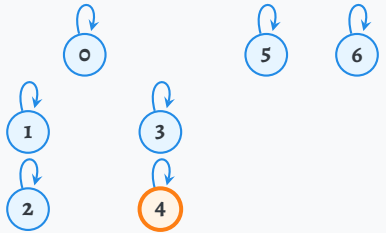
`find(4)`

- Read `parent[4] = 4`.
- The node is its own parent: return 4.

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	6

DSU Forest



`union`: Find Both Roots, Then Link

Pseudocode

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union(first, second):
    rootA ← find(first)
    rootB ← find(second)
    if rootA ≠ rootB then
        parent[rootB] ← rootA
```

`union`: Find Both Roots, Then Link

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1 union(first, second):
2     rootA ← find(first)
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4     if rootA ≠ rootB then
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```

Parent array

index	0	1	2	3	4	5	6
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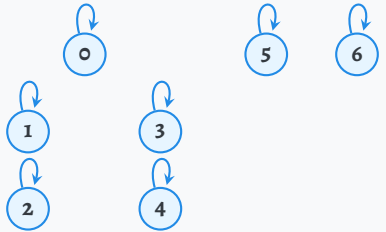
DSU Forest



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2     rootA ← find(first)
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4     if rootA ≠ rootB then
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```

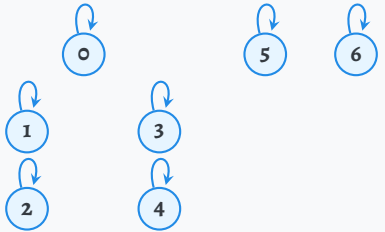
Operations

`union(1, 2)`

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	6

DSU Forest



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```

Operations

`union(1, 2)`

`find(1):parent[1] = 1` $\Rightarrow$ rootA $= 1$

Parent array

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parent	0	1	2	3	4	5	6

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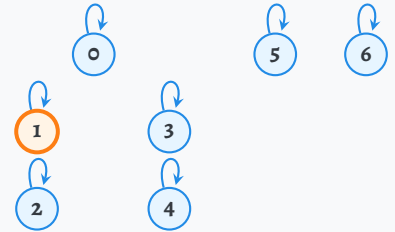

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```

Operations

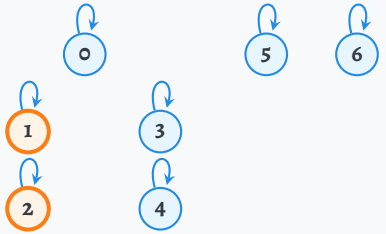
`union(1, 2)`

`find(1):parent[1] = 1` $\Rightarrow$ rootA $= 1$
`find(2):parent[2] = 2` $\Rightarrow$ rootB $= 2$

Parent array

index	0	1	2	3	4	5	6
parent	0	1	2	3	4	5	6

DSU Forest



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4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

`union(1, 2)` `parent[2] ← 1`

`find(1)`: `parent[1] = 1` ⇒ rootA = 1
`find(2)`: `parent[2] = 2` ⇒ rootB = 2
Different roots: attach 2 under 1.

Parent array

index	0	1	2	3	4	5	6
parent	0	1	1	3	4	5	6

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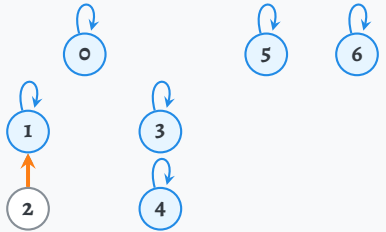

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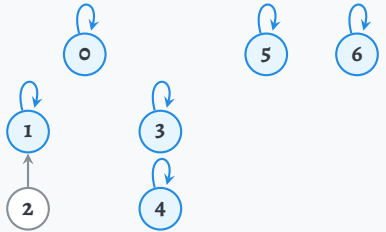
Operations

```
union(1, 2)  parent[2] ← 1
union(0, 1)
```

Parent array

index	0	1	2	3	4	5	6
parent	0	1	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

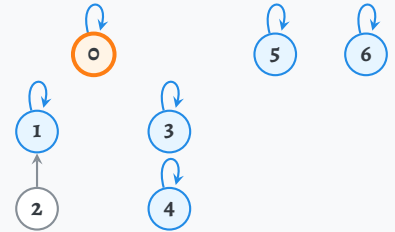
union(1, 2) parent[2] ← 1
union(0, 1)

find(0):parent[0] = 0 ⇒ rootA = 0

Parent array

index	0	1	2	3	4	5	6
parent	0	1	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

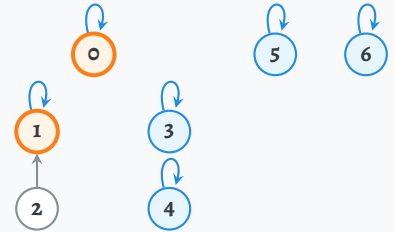
`union(1, 2)` parent[2] ← 1
`union(0, 1)`

`find(0)`: parent[0] = 0 ⇒ rootA = 0
`find(1)`: parent[1] = 1 ⇒ rootB = 1

Parent array

index	0	1	2	3	4	5	6
parent	0	1	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
```

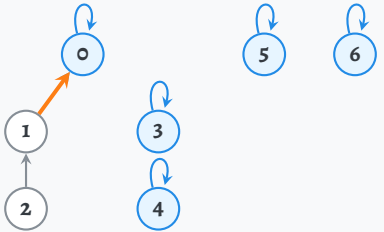
```
find(0):parent[0] = 0 ⇒ rootA = 0
find(1):parent[1] = 1 ⇒ rootB = 1
```

Different roots: attach 1 under 0.

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

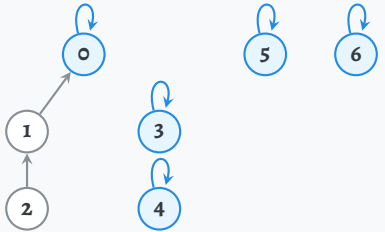
Operations

```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)
```

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

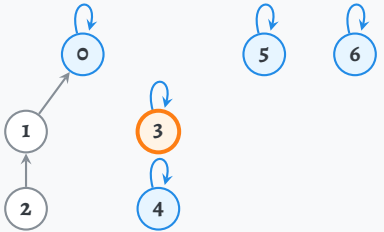
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)
```

find(3):parent[3] = 3 ⇒ rootA = 3

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

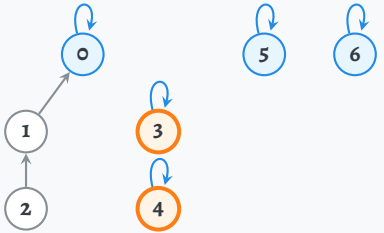
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)
```

```
find(3):parent[3] = 3 ⇒ rootA = 3
find(4):parent[4] = 4 ⇒ rootB = 4
```

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	4	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

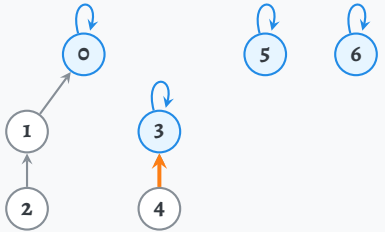
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
```

`find(3)`: `parent[3] = 3` ⇒ rootA = 3
`find(4)`: `parent[4] = 4` ⇒ rootB = 4
Different roots: attach 4 under 3.

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

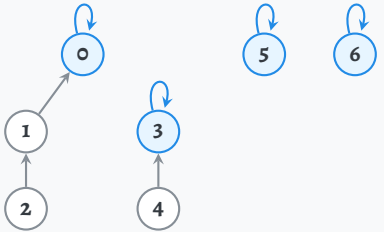
Operations

```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)
```

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

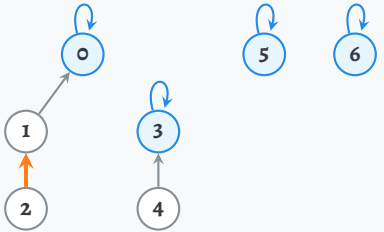
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)

find(2): 2 → 1
```

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

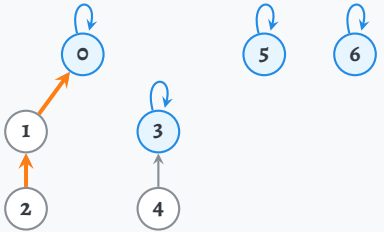
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)

find(2): 2 → 1 → 0
```

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

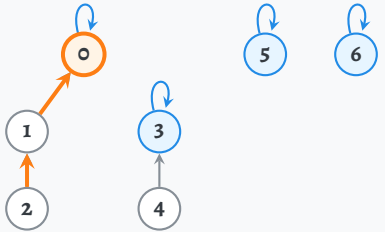
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)
```

find(2): 2 → 1 → 0; parent[0] = 0 ⇒ rootA = 0

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

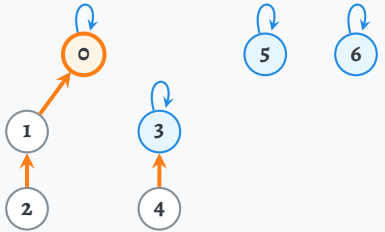
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)
```

find(2): $2 \rightarrow 1 \rightarrow 0$; parent[0] = 0 $\Rightarrow$ rootA = 0
find(4): $4 \rightarrow 3$

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

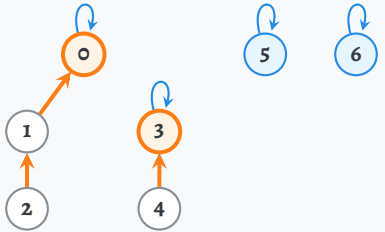
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)
```

find(2): 2 → 1 → 0; parent[0] = 0 ⇒ rootA = 0
find(4): 4 → 3; parent[3] = 3 ⇒ rootB = 3

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	3	3	5	6

DSU Forest



union: Find Both Roots, Then Link

Pseudocode

```
1 union(first, second):
2     rootA ← find(first)
3     rootB ← find(second)
4     if rootA ≠ rootB then
5         parent[rootB] ← rootA
```

Operations

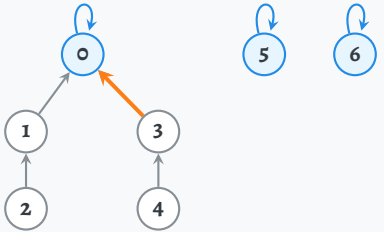
```
union(1, 2)  parent[2] ← 1
union(0, 1)  parent[1] ← 0
union(3, 4)  parent[4] ← 3
union(2, 4)  parent[3] ← 0
```

`find(2)`: $2 \rightarrow 1 \rightarrow 0$; `parent[0] = 0` $\Rightarrow$ rootA $= 0$
`find(4)`: $4 \rightarrow 3$; `parent[3] = 3` $\Rightarrow$ rootB $= 3$
Different roots: attach 3 under 0, not 4 under 2.

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	0	3	5	6

DSU Forest



Path Compression

Naive `find`: The Same Work Again

Pseudocode

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
```

Parent array

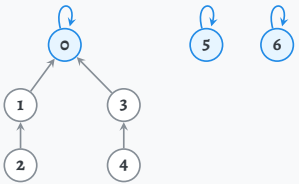
index	0	1	2	3	4	5	6
parent	0	0	1	0	3	5	6

Exactly where the previous `union` left us.

Operations

- Call `find(2)`:

DSU Forest



Naive `find`: The Same Work Again

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      return find(parent[node])
```

Parent array

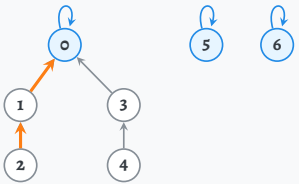
index	0	1	2	3	4	5	6
parent	0	0	1	0	3	5	6

Exactly where the previous `union` left us.

Operations

- Call `find(2)`: $2 \to 1 \to 0$.

DSU Forest



Naive `find`: The Same Work Again

Pseudocode

```
find(node):
    if parent[node] = node then
        return node
    return find(parent[node])
```

Parent array

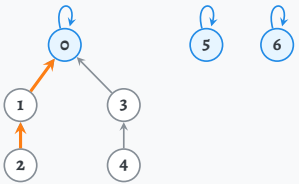
index	0	1	2	3	4	5	6
parent	0	0	1	0	3	5	6

Exactly where the previous `union` left us.

Operations

- Call `find(2)`: $2 \to 1 \to 0$.
- Call it again: the same two links!

DSU Forest



Naive `find`: The Same Work Again

Pseudocode

```
find(node):
    if parent[node] = node then
        return node
    return find(parent[node])
```

Parent array

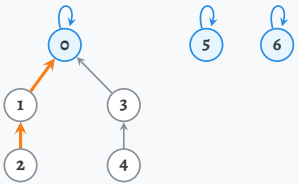
index	0	1	2	3	4	5	6
parent	0	0	1	0	3	5	6

Exactly where the previous `union` left us.

Operations

- Call `find(2)`: $2 \to 1 \to 0$.
- Call it again: the same two links!
- The answer is returned, but no parent is updated.

DSU Forest



Naive `find`: The Same Work Again

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      return find(parent[node])
```

Parent array

index	0	1	2	3	4	5	6
parent	0	0	1	0	3	5	6

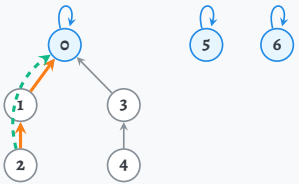
Exactly where the previous `union` left us.

Operations

- Call `find(2)`: $2 \to 1 \to 0$.
- Call it again: the same two links!
- The answer is returned, but no parent is updated.

Can we remember the shortcut to 0?

DSU Forest



Naive Union Can Build a Chain

A fresh worst-case example

Start again with seven singleton sets.
Our union rule: `parent[rootB]` ← `rootA`.

- `union(1,0)`, `union(2,1)`, `union(3,2)`, ..., `union(6,5)`.

Naive Union Can Build a Chain

A fresh worst-case example

Start again with seven singleton sets.
Our union rule: `parent[rootB]` ← `rootA`.

- `union(1,0)`, `union(2,1)`, `union(3,2)`, ..., `union(6,5)`.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

Naive Union Can Build a Chain

A fresh worst-case example

Start again with seven singleton sets.
Our union rule: `parent[rootB]` ← `rootA`.

- `union(1,0)`, `union(2,1)`, `union(3,2)`, ..., `union(6,5)`.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

DSU Forest

Arrows point to parents; 6 is the root.

Naive Union Can Build a Chain

A fresh worst-case example

Start again with seven singleton sets.
Our union rule: `parent[rootB]` ← `rootA`.

- `union(1,0)`, `union(2,1)`, `union(3,2)`, ..., `union(6,5)`.
- `find(0)` follows six links.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

DSU Forest

Arrows point to parents; 6 is the root.

Naive Union Can Build a Chain

A fresh worst-case example

Start again with seven singleton sets.
Our union rule: `parent[rootB]` ← `rootA`.

- `union(1,0)`, `union(2,1)`, `union(3,2)`, ..., `union(6,5)`.
- `find(0)` follows six links.
- For n elements: up to $n - 1$ links, so one `find` can take $\Theta(n)$.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

DSU Forest

Arrows point to parents; 6 is the root.

Naive `union` also calls `find`, so it can take $\Theta(n)$ too.

Path Compression: Save the Root on the Way Back

Before: only return the root

```
find(node):
    if parent[node] = node then
        return node
    return find(parent[node])
```

- Follow parents until reaching the root, exactly as before.

Path Compression: Save the Root on the Way Back

Before: only return the root

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
```

After: also update each visited parent

```
1 find(node):
2     if parent[node] = node then
3         return node
4     parent[node] ← find(parent[node])
5     return parent[node]
```

- Follow parents until reaching the root, exactly as before.
- As recursion returns, make **every visited node** point directly to that root.

Path Compression: Save the Root on the Way Back

Before: only return the root

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
```

After: also update each visited parent

```
1 find(node):
2     if parent[node] = node then
3         return node
4     parent[node] ← find(parent[node])
5     return parent[node]
```

- Follow parents until reaching the root, exactly as before.
- As recursion returns, make **every visited node** point directly to that root.
- Same set. Same representative. **Shorter paths.**

Path Compression: Save the Root on the Way Back

Before: only return the root

```
1 find(node):
2     if parent[node] = node then
3         return node
4     return find(parent[node])
```

After: also update each visited parent

```
1 find(node):
2     if parent[node] = node then
3         return node
4     parent[node] ← find(parent[node])
5     return parent[node]
```

- Follow parents until reaching the root, exactly as before.
- As recursion returns, make **every visited node** point directly to that root.
- Same set. Same representative. **Shorter paths.**

Find the root, then remember it.
No change to `make_set` or the `union` pseudocode.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: 0 → 1 .

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: $0 \to 1 \to 2$.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
find(node):
    if parent[node] = node then
        return node
    parent[node] ← find(parent[node])
    return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: $0 \to 1 \to 2 \to 3$.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: $0 \to 1 \to 2 \to 3 \to 4$.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	**4**	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: $0 \to 1 \to 2 \to 3 \to 4 \to 5$.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
find(node):
    if parent[node] = node then
        return node
    parent[node] ← find(parent[node])
    return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 6$.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Follow the original parent pointers.

Simulating `find(0)`: First Reach the Root

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

- Begin with the chain: call `find(0)`.
- Calls: $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 6$.
- `parent[6] = 6`: return 6.

Now the suspended calls can update their parents.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

No assignments yet: recursive calls are still pending.

DSU Forest

Root found: return 6.

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

`find(6)` returns 6.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
find(node):
    if parent[node] = node then
        return node
    parent[node] ← find(parent[node])
    return parent[node]
```

Operations

`parent[5]` ← 6; return 6.
Already points to the root: no visible change.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	5	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

`parent[4]` ← 6; return 6.
Replace 4 → 5 with 4 → 6.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	4	6	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

`parent[3]` ← 6; return 6.
Replace 3 → 4 with 3 → 6.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	3	6	6	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

`parent[2]` ← 6; return 6.
Replace 2 → 3 with 2 → 6.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	1	2	6	6	6	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

`parent[1]` ← 6; return 6.
Replace 1 → 2 with 1 → 6.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	1	6	6	6	6	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

Returning: Every Visited Node Gets the Shortcut

Pseudocode

```
1  find(node):
2      if parent[node] = node then
3          return node
4      parent[node] ← find(parent[node])
5      return parent[node]
```

Operations

`parent[0]` ← 6; return 6.
Done: every non-root now points to 6.

Return order: 6, 5, 4, 3, 2, 1, 0.

Parent array

index	0	1	2	3	4	5	6
parent	6	6	6	6	6	6	6

Orange cell: active parent entry.
Teal arrow: a saved shortcut.

DSU Forest

The Next `find(0)`: Six Links Become One

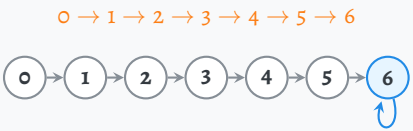
Before compression

$0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 6$

First search: six parent links.
Without compression, every repeat costs six.

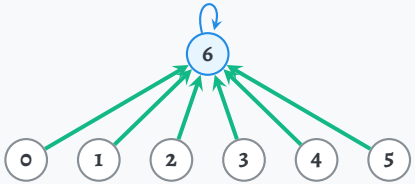
The Next `find(0)`: Six Links Become One

Before compression



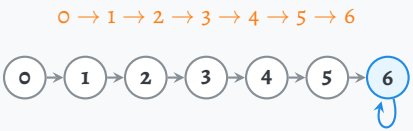
First search: six parent links.
Without compression, every repeat costs six.

After compression: the same nodes rearranged



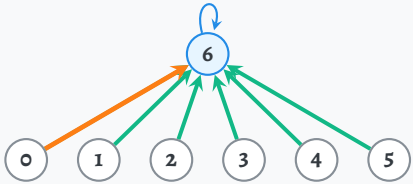
The Next `find(0)`: Six Links Become One

Before compression



First search: six parent links.
Without compression, every repeat costs six.

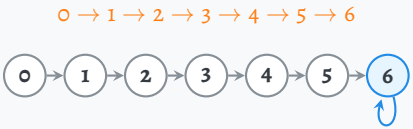
After compression: the same nodes rearranged



Next search: one parent link, 0 → 6.
Other visited nodes benefit too.

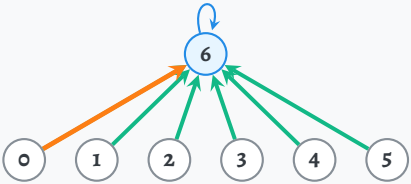
The Next `find(0)`: Six Links Become One

Before compression



First search: six parent links.
Without compression, every repeat costs six.

After compression: the same nodes rearranged



Next search: one parent link, $0 \to 6$.
Other visited nodes benefit too.

For $k \geqslant 1$ calls to `find(0)`, with **no intervening unions**:
naive: $6k$ parent-link traversals vs. compression: $6 + (k - 1)$.